Software Requirements Specification

for

Random Access Market

Version 1.0 approved

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1. Introduction

1.1 Purpose

Random Access Market is an e-commerce platform designed to provide customers with an online consumer shopping experience. This document is designed to be digestible to the client in terms of what technical specifications, requirements, scope, and the design of the Random Access Market are outlined.

1.2 Document Conventions

This document uses a black Arial font. Top-level section headings are typed in a bold 18pt size and are numbered sequentially (1, 2, 3, etc.). Lower-level sections are typed in a 14pt bold size, and are numbered through decimal points that reflect their placement below a specific top-level section (i.e., sections under 1 are numbered “1.1, 1.2, 1.3” etc.).

Descriptions under designated features in Section 3 are specifically written in an italicized 11pt black Arial font. Priority labels are written in a 12pt italicized and bold black Arial font. Stimulus/response-sequences are displayed in a 12pt blue Arial font. Glossary words are written in a bold 12pt black Arial font. Information is otherwise typically conveyed through a black 12pt Arial font.

Page numbers are specified in a courier 10pt black font.

1.3 Intended Audience and Reading

The intended audience of the Random Access Market SRS document is the platform’s administrators and maintenance developers, along with invested shareholders and supervisorial interests.

1.4 Product Scope

The Random Access Market is an e-commerce platform intended to enable users to register accounts for engaging in buying and selling transactions from and to other users.

1.5 References

- Svelte
  - https://svelte.dev/

- MariaDB
  https://mariadb.org/
- Prisma
  https://www.prisma.io/
- ExpressJS
  https://expressjs.com/

2. Overall Description

2.1 Product Perspective

Random Access Market is a new and self-contained product that does not exist within any other pre-existing product line. It is a web application that will facilitate transactions between seller users and buyer users.

2.2 Product Functions

- The user must be able to log into the application using username and password credentials.
- The platform must support the self-assignment of buyer and seller roles, and the delegation of the administrator role.
- Allow administrators to moderate the platform to ensure terms of service and legal compliance.
- The user will be able to browse the platform for products
- The product must be able to store username and password credentials in a relational database.

2.3 User Classes and Characteristics

Three user types are expected to use the application: buyers, sellers, and administrators.

- Buyers are users who will use the application to search for and purchase products.
- Sellers are users who will submit products to the application for sale.
- Administrators are users who will review and approve new users and new products on the web service.

2.4 Operating Environment

The software will operate as a web application and will be compatible with any operating system the user uses, provided he/she has a web browser to access the website hosting the software. The software will operate most optimally for Firefox and Chromium-based browsers.

2.5 Design and Implementation Constraints

2.6 Design:

- Frontend: SvelteJS

- Backend: ExpressJS, Prisma, MariaDB

2.7 Constraints:

- HB1126 (Walker Montgomery Protecting Children Online Act) and similar age and identity-verification laws in various jurisdictions could affect platform availability or feature implementation.
  - Note: The bill makes an explicit exemption of platforms primarily serving a commerce purpose, however
- The platform will be optimized primarily for Firefox and Chromium-based browsers, as these constitute the vast majority of used-browsers and further attempts to increase browser accessibility would yield diminishing returns.
- The platform will primarily target an English-speaking audience and only support English due to a current lack of human resources and capital to moderate non-English content.

3. System Features

3.1 Account Registration

3.1.1 Description and Priority

High Priority: 9 (Essential)

The system provides a way for new users to create an account via submitting required registration information. The system validates the information, ensures the lack of duplicate accounts, and confirms successful registration.

3.1.2 Stimulus/Response Sequences

[user selects name/password entry area → system receives request; indicates entry area is selected and awaiting input]

[user inputs data in name/password entry areas → system accepts input; stores data in entry area data, displays data currently stored in name entry area; displays censor characters representing data currently stored in password entry area]

[user selects confirmation area → system receives request; creates account with input data]

3.1.3 Functional Requirements

FREQ-1: The system shall allow a guest user to submit a registration request containing a unique username and password.

FREQ-2: The system shall send and display to the user an error message that indicates the username credential is invalid if the provided username credential is already registered.

FREQ-3: The system will send and display to the user an error message upon the entry of credentials that do not meet platform-defined format specifications (see 4.3.4)

FREQ-4: The system shall create a new user account with the selected role upon successful registration.
FREQ-5: The system shall display a confirmation message upon successful registration.
FREQ-6: The system shall query an admin for approval of new user accounts.

3.2 User Authentication

3.2.1 Description and Priority

High Priority: 9 (Essential)

The process for registered users to securely log in and gain access to their personalized account features. This enables registered users to access the system as non-guest users.

3.2.2 Stimulus/Response Sequences

[user selects name/password entry area → system receives request; indicates entry area is selected and awaiting input]

[user inputs data in name/password entry areas → system accepts input; stores data in entry area data, displays data currently stored in name entry area; displays censor characters representing data currently stored in password entry area]

[user selects confirmation area → system receives request; searches for existing account with data matching input; displays error or home page]

3.2.3 Functional Requirements

FREQ-7: The system shall allow registered users to log in using correct credentials
FREQ-8: The system shall display an error message if the user enters incorrect login credentials
FREQ-9: The system shall deny authentication when invalid credentials are provided
FREQ-10: The system shall redirect the user to the home page upon successful login.
FREQ-11: The system shall assign an authentication token on successful login
FREQ-12: The system shall restore user data upon successful login if there is data to restore

3.3 Home Page Display

3.3.1 Description and Priority

High priority: 9 (Essential)

The Home Page will provide a usable, visual interface that provides the user with immediate access to key features of the system (such as Account Management, Order Tracking, and Order History). This system's usability is contingent on the home page display feature.

3.3.2 Stimulus/Response Sequences

[user selects search index entry area → system receives request; indicates search area is selected and awaiting input]

[user inputs data in search entry → system accepts input; stores data in entry area data, displays data currently stored in entry area]

[user selects search action → system receives request; retrieves database entries filtered by input; creates a list of results; displays the list of results]

[user selects link to alternate page → system receives request; retrieves alternate page data; loads alternate page data; displays alternate page]

[user selects home page icon → system receives request; retrieves home page data; loads home page; displays home page]

3.3.3 Functional Requirements

FREQ-13: The system shall accept user input for option selection and textbox entries.

FREQ-14: The system shall display categories for the user to select for filtered searching.

FREQ-15: The system shall display context specific options for logged in accounts, such as creating a new product listing, viewing cart list, viewing banned accounts listing, etc.

FREQ-16: The system shall not display context specific options for accounts that do not meet role requirements for specified options (admin options unavailable for buyers and sellers, etc).

FREQ-17: The system shall offer a selectable option to log out that signals other components to clear their session data

3.4 Product Search

3.4.1 Description and Priority

High Priority: 9 (Essential)

Unique account object which provides a user with the ability to browse product listings and consider them for purchase.

3.4.2 Stimulus/Response Sequences

[user selects search index entry area → system receives request; indicates search area is selected and awaiting input]

[user inputs data in search entry → system accepts input; stores data in entry area data, displays data currently stored in entry area]

[user selects search action → system receives request; retrieves database entries filtered by input; creates a list of results; displays the list of results]

[user selects link to alternate page → system receives request; retrieves alternate page data; loads alternate page data; displays alternate page]

[user selects home page icon → system receives request; retrieves home page data; loads home page; displays home page]

3.4.3 Functional Requirements

FREQ-18: The system shall allow users to search for products by entering keywords into a search bar.
FREQ-19: The system shall provide filters to narrow search results by product category.
FREQ-20: The system shall provide filters to narrow search results by price range.

3.5 Product Viewing

3.5.1 Description and Priority

High Priority: 9 (Essential)

Unique account object which provides a user with ability to browse product listings and consider them for purchase.

3.5.2 Stimulus/Response Sequences

[user selects product entry → system receives request; retrieves product details; loads product details; displays product details page]

3.5.3 Functional Requirements

FREQ-21: The system shall display the price of the product.
FREQ-22: The system shall display the product description and specifications.
FREQ-23: The system shall display one or more images associated with the product.
FREQ-24: The system shall allow buyers to add products to their cart.

3.6 Compare products

3.6.1 Description and Priority

High Priority: 9 (Essential)

The system provides a visual interface for the buyer to view details of selected products for comparison.

3.6.2 Stimulus/Response Sequences

[potential user story: As a buyer I want to view the differences of products I am considering purchasing to assist in my decision]

[user selects compare product option → system receives request; retrieves product details; adds product to buyer comparison list; displays product details in comparison view]

[user has not selected second product for comparison → system verifies no second product selected; displays request for buyer to select second product for comparison]

3.6.3 Functional Requirements

FREQ-25: The system shall display the price of the product.
FREQ-26: The system shall display the product description and specifications.
FREQ-27: The system shall display one or more images associated with the product.
FREQ-28: The system shall arrange the above in an organized manner that allows a buyer to view each element of two selected products for comparison

3.7 Cart/Checkout

3.7.1 Description and Priority

High Priority: 9 (Essential)

The system provides a visual interface for the buyer to view and edit product selections for purchase before continuing with purchase.

3.7.2 Stimulus/Response Sequences

[buyer requests to view cart → system receives request; retrieves cart list; displays products in list]

[buyer requests product removal from cart → system receives request; removes product from cart list; updates displayed products]

[buyer requests to add product to cart → system receives request; creates product entry; adds entry to cart list; updates displayed products]

[buyer requests to continue with cart to checkout → system receives request; retrieves individual product cost for each in cart; sums costs; applies relevant extra costs; displays total]

3.7.3 Functional Requirements

FREQ-29: The system shall allow buyers to view the products stored in their cart.
FREQ-30: The system shall allow buyers to remove products from their cart.
FREQ-31: The system shall allow buyers to add products to their cart.
FREQ-32: The system shall allow buyers to initiate a purchase transaction for available products.
FREQ-33: The system shall allow buyers to review order details before confirming their purchase.
FREQ-34: The system shall prevent buyers from adding out-of-stock products to their cart.
FREQ-35: The system shall prevent buyers from attempting to continue to checkout with no products in their cart.
FREQ-36: The system shall keep products in buyer's cart until logout detected.

3.8 Seller Inventory Management

3.8.1 Description and Priority

High Priority: 9 (Essential)

The system provides a way for sellers to update product listings and stock information. This feature allows sellers to maintain accurate inventory for items offered on the platform.

3.8.2 Stimulus/Response Sequences

[seller requests to view inventory → system receives request; retrieves seller inventory list; displays products in list]

[seller requests to remove product from account inventory → system receives request; removes product from seller inventory list; updates displayed products]

[seller requests to add product to account inventory → system receives request; creates product entry; adds entry to seller inventory; updates displayed products]

[buyer purchases seller's listing → system conducts transaction; removes products from seller's inventory; sends buyer's funds to seller's registered financial account]

[buyer returns purchase → system searches for order containing returned products and buyer's payment method; retrieves payment method used in purchase; retrieves products; returns products to seller's inventory; returns funds to buyer's financial account]

3.8.3 Functional Requirements

FREQ-37: The system shall allow sellers to view their inventory.
FREQ-38: The system shall allow sellers to remove products from their inventory.
FREQ-39: The system shall allow sellers to add products to their inventory.
FREQ-40: The system shall allow sellers to create product listings.
FREQ-41: The system shall allow sellers to delete product listings.
FREQ-42: The system shall allow sellers to edit product listings.
FREQ-43: The system shall allow sellers to update product stock.
FREQ-44: The system shall automatically update seller inventory upon a buyer purchasing their stock.
FREQ-45: The system shall not allow sellers to add negative amounts of product.
FREQ-46: The system shall not allow sellers to remove more product than what is currently available in inventory.
FREQ-47: The system shall not allow sellers to create empty listings.
FREQ-48: The system shall allow buyers to return products they have purchased.
FREQ-49: The system shall automatically add products that have been returned to seller's inventory.
FREQ-50: The system shall automatically send funds from sales to seller's initialized financial account.
FREQ-51: The system shall automatically send funds from returns to buyer's initialized financial account.

3.9 Order Tracking

3.9.1 Description and Priority

Mid Priority: 5 (Useful)

The system allows the buyer to track active orders and their status.

3.9.2 Stimulus/Response Sequences

[buyer requests to view ongoing orders → system receives request; retrieves page data; loads page; displays page]

[buyer selects ongoing order to view → system receives request; retrieves order details data; loads details page; displays details page]

[buyer selects cancel order → system receives request; retrieves products listed in order; adds products back to offering seller's inventory; refunds buyer]

3.9.3 Functional Requirements

FREQ-52: The system shall allow users to view orders that are ongoing.
FREQ-53: The system shall display the current status of an ongoing order.
FREQ-54: The system shall allow buyers to view tracking info for orders being delivered / in transit.
FREQ-55: The system shall update order status as it progresses.
FREQ-56: The system shall automatically send orders marked as completed to order history.
FREQ-57: The system shall automatically delete ongoing order entries after they are sent to order history.
FREQ-58: The system shall allow buyers to cancel orders before completion.
FREQ-59: The system shall automatically return products to the inventory of its original seller upon order cancellation.
FREQ-60: The system shall automatically refund the buyer's purchase upon order cancellation.

3.10 Order History

3.10.1 Description and Priority

Mid Priority: 5 (Useful)

The system provides the buyer a means to view records of previously-conducted orders.

3.10.2 Stimulus/Response Sequences

[buyer selects Order History → system displays reverse-chronological list of purchase records]

[buyer enters string in Order History search field → system executes database search query for orders table filtered by buyer id, system refreshes page to display yielded records]

[buyer enters string in Order History search field → system executes database search query for orders table filtered by buyer id, system refreshes page to display “No orders shown” if no match]

3.10.3 Functional Requirements

FREQ-61: The system shall maintain a record of completed orders.
FREQ-62: The system shall allow buyers to view their completed purchases in reverse-chronological order.
FREQ-63: The system shall display the product name, the product thumbnail image, and the seller username with each purchase.
FREQ-64: The system shall provide the user with a “reorder” action for each product that adds the product to the buyer’s cart.
FREQ-65: The system shall provide a search field that allows the buyer to find records by product or seller name.

3.11 Buyer Account Management

3.11.1 Description and Priority

High Priority: 8 (Essential)

Page that acts as a personal hub of useful information for buyer users

3.11.2 Stimulus/Response Sequences

[buyer selects valid file from device → system checks file extension for .png or .jpg; system checks file size is under limit; system saves file to server’s image directory]

[buyer selects file of an invalid format → system checks file extension for .png or .jpg; system displays error message: “Invalid file type, upload .png or .jpg”]

[buyer selects file exceeding file size limit → system checks file extension for .png or .jpg; system checks file size is under limit, system displays error message: “File exceeds 2MB maximum size”]

[buyer submits completed payment credentials form → system validates payment format; system generates unique payment token id and stores in database table]

[buyer submits new display name → system receives input; system checks against database for duplicate display name entries; system overwrites current record for user display name]

[buyer submits occupied display name → system receives input; system checks against database for duplicate name entries; system displays “display name taken” error message]

3.11.3 Functional Requirements

FREQ-66: The system shall enable the buyer to provide shipping address details, including name, street address, city, state/region, postal number/zip, and country.

FREQ-67: The system shall allow the buyer to update their profile picture through uploading a .png or .jpg file.

FREQ-68: The system shall allow the buyer to input and save credit/payment card or bank details.

FREQ-69: The system shall allow the user to input and save a non-duplicate display name.

3.12 Seller Account Management

3.12.1 Description and Priority

High Priority : 8 (Essential)

The system will allow seller accounts to display their products on their own page that is accessible to all buyer accounts

3.12.2 Stimulus/Response Sequences

[seller selects valid file from device → system checks file extension for .png or .jpg; system checks file size is under limit; system saves file to server’s image directory]

[seller selects file of an invalid format → system checks file extension for .png or .jpg; system displays error message: “Invalid file type, upload .png or .jpg”]

[seller selects file exceeding file size limit → system checks file extension for .png or .jpg; system checks file size is under limit, system displays error message: “File exceeds maximum size”]

[seller submits completed bank credentials form → system checks form completion; system generates id to store in database]

[buyer submits payment (credit or banking card) for product → system validates payment format; system validates payment availability; system withdraws funds corresponding to total product price from buyer-provided payment medium; system transfers funds to seller account virtual wallet]

[buyer submits payment (credit or banking card) of invalid format for product → system checks payment format; system rejects payment and displays “invalid payment format” message to buyer]

[buyer submits payment (credit or banking card) with insufficient funds for product → system validates payment format; system checks payment availability; system rejects payment and displays “insufficient funds” error]

[seller submits virtual wallet withdrawal request for amount <= virtual wallet funds into selected bank → system validates amount is <= virtual wallet funds; system removes requested amount of funds from virtual wallet; system sends requested amount of funds to seller’s bank account; system logs record with transaction id in database]

[seller submits virtual wallet withdrawal request for amount > virtual wallet funds into selected bank → system checks amount is <= virtual wallet funds; system halts initialization of transaction process and displays “insufficient amount of funds”]

3.12.3 Functional Requirements

FREQ-70: The system shall allow sellers to update their profile image through uploading a .jpg or .png image.
FREQ-71: The system shall allow sellers to update their store text description.
FREQ-70: The system shall allow sellers to update their profile name through a text field.
FREQ-71: The system shall allow the seller to input and update their contact information.
FREQ-72: The system shall allow the seller to input and save banking information.
FREQ-73: The system shall allow the seller to receive payment from purchased products into a virtual wallet.
FREQ-74: The system shall allow the seller to withdraw funds from virtual wallet into a selected bank account.
FREQ-75: The system shall deny seller accounts with an unapproved status from selling or listing products.
FREQ-76: The system shall deny seller accounts with a banned status from selling or listing products.

3.13 Administration

3.13.1 Description and Priority

High Priority: 9 (Essential)

The system will allow administrator accounts to ban seller and buyer accounts. This will ensure the platform is adequately moderated and that fraudulent or illegal behaviors are discouraged.

3.13.2 Stimulus/Response Sequences

[administrator selects ban action for seller user → system updates seller status from active to banned; system checks for active seller listings; system hides all of seller’s product listings; system logs unique audit log id into database; system denies further authentication of seller user]

[administrator selects ban action for buyer user → system updates buyer status from active to banned; system checks buyer user's pending orders; system cancels buyer user's pending orders; system undoes inventory depreciation; system logs unique audit log id into database; system denies further authentication of buyer user]

[super-administrator selects admin status to apply to user → system updates user status to administrator; system logs status change into database with unique audit log id]

[administrator selects status to approved for seller user → system updates seller status to approved; system logs approval in database with unique audit log id]

[administrator selects status to approved for buyer user → system updates buyer status to approved; system logs approval in database with unique audit log id]

[super-administrator selects non-administrator status for admin → system updates administrator status to non-administrator status; system logs status change into database with unique audit log id]

[super-administrator selects ban status for administrator → system updates administrator status to non-administrator status; system updates user status to banned; system logs unique audit log id into database; system denies further authentication to user

3.13.3 Functional Requirements

FREQ-77: The system shall provide an administrative panel that allows updating the status of a seller user account between unapproved, approved, and banned.

FREQ-78: The system shall provide an administrative panel that allows updating the status of a buyer account between unapproved, approved, and banned.

FREQ-79: The system shall provide the super-administrator with an administrative panel that allows updating the status of any other user between non-administrator and administrator.

FREQ-80: The system shall provide the super-administrator with all administrator permissions.

FREQ-81: The system shall hide all product listings from seller accounts with ban status.

FREQ-82: The system shall display a notification of the registered account's banned status upon an attempt at authentication.

FREQ-83: The system shall provide a filter to view seller users by unapproved, approved, or banned status.

FREQ-84: The system shall provide a filter to view buyer users by unapproved, approved, or banned status.
FREQ-85: The system shall provide a filter to view users by time of registration.
FREQ-86: The system shall deny authentication to buyer users with ban status.

3.14 View Random Product

3.14.1 Description and Priority

Mid Priority: 4 (Useful)

The system will offer an option for the buyer to request a randomly selected product to be shown to them.

3.14.2 Stimulus/Response Sequences

[buyer selects option to view random product → system receives request; seeds random function; generates random value; uses value to search for product; retrieves product details; loads product details; displays product details page]

3.14.3 Functional Requirements

FREQ-87: The system shall offer the buyer a randomly selected product per request.

4. Other Nonfunctional Requirements

4.1 Performance Requirements

4.1.1 The system shall support multiple users at once without significant performance decline
4.1.2 The system shall load pages within two seconds.
4.1.3 The system shall return search results within less than a second.
4.1.4 The system shall limit image uploads to a maximum of 2MB to optimize storage and load times.

4.2 Safety Requirements

4.2.1 The system shall prevent unintentional loss of user data
4.2.2 The system shall provide system failure recovery mechanisms
4.2.3 The system shall ensure that transactions are reliably processed

4.3 Security Requirements

4.3.1 The system shall require authentication for user account access
4.3.2 The system shall verify user credentials before granting access to administrative features
4.3.3 The system shall only collect and store personally identifiable information (PII) that is necessary for the legitimate commercial operations of the platform
4.3.4 The system shall enforce a password policy requiring a minimum of 8 characters, including a mix of uppercase letters, lowercase letters, numbers, and symbols.
4.3.5 The system shall only allow the uploading of specific image file types
4.3.6 The system shall set the status of seller accounts to unverified by default.

4.4 Software Quality Attributes

4.4.1 The system shall provide a user-friendly interface
4.4.2 The system shall be maintainable to support future updates
4.4.3 The system shall be scalable to accommodate growth
4.4.4 The system shall be designed to support high uptime

5. Other Requirements

5.1 The system shall verify the inventory level of products within less than a second.

Appendix A: Glossary

- **Actor** - a person interacting with the software application.
- **User** - actor with credentials and an account
- **Buyer**- user with the buyer role
- **Seller**- user with the seller role
- **Admin** - user with the administrator role.
- **Super-Admin** - user with admin role permissions who can delegate the admin role to users.